

Oscar Liang

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EDUCATION

University of Toronto - BAsC. in Engineering Science (Robotics Specialization) + PEY

Sept 2024 - May 2029

Minor in Artificial Intelligence, EV Design Certificate, Engineering Business Certificate

EXPERIENCES

New York University Shanghai

May 2026 - Present

Student Research Assistant

- Architected an end-to-end VLA pipeline for a wheelchair-mounted robotic arm, enabling assistive manipulation for individuals with limited upper-limb mobility.
- Fine-tuned and deployed a SmolVLA action-generation module that converts dual-camera tray workspace views into closed-loop motor trajectories for successful precision cup grasping and mouth-pose delivery.
- Illustrated the multi-stage system architecture diagrams and co-authored the peer-reviewed poster detailing pipeline designs.

HighGreat Technology Co., Ltd

Summer 2025 & 2026

Project Intern

- Programmed keyframe trajectories and dynamic lighting choreography across Fylo, HULA, and EMO drone platforms, executing synchronized drone swarm performances for commercial shows with 2,000+ spectators per show across China.
- Leveraged 3D simulation pre-visualization environments to test, debug, and optimize critical anti-collision parameters, validating flight safety and pathing prior to physical hardware deployment.

University of Toronto Club of Biomedical Engineering

Jan 2025 - Present

Conference + Sponsorship Director

- Directed a 200+ attendee biomedical engineering conference, pioneering a 3-track structure (Research, Industry, Innovation) in delivering targeted career insights and expanding external partner engagement.
- Spearheaded high-volume procurement campaigns to 500+ sponsors, targeting medical industry leaders and academic researchers; received funding and equipment valued at over \$1,500 CAD.

York Region District School Board

Jun 2023 - Sept 2024

International Student Ambassador

- Presented key insights at International Student Orientation panels for 800+ attendees, sharing experiences to ease transitions.
- Raised over \$2,000 CAD and 475+ lbs of donations, doubling the set goal of \$1,000 CAD, for the Richmond Hill Food Bank by communicating and managing sponsorships with the Canadian Asian Roots Collective.

PROJECTS

BracketBot Autonomous Butler

Sept 2026

- Engineered an ArUco marker localization pipeline on BracketBot using OpenCV's PnP pose solver and fisheye camera calibration, incorporating pre-grasp wrist-camera checks to ensure reliable manipulation.
- Architected the visual-servo control loop and finite-state-machine logic governing BracketBot's tray scooping and delivery, executing a zero-teleoperation live demo to secure 2nd Place at the Battle of the Schools Hackathon.

Physics Informed Neural Network on Harmonic Oscillators

Apr 2025

- Engineered three Physics-Informed Neural Network (PINN) architectures (Vanilla, Fourier Feature, SIREN) in PyTorch to model damped harmonic motion, tuning loss weights, activation functions, and learning rates; revealed that the Fourier Feature architecture achieved 100x higher accuracy than baseline models, while SIREN exhibited the fastest training convergence.

Computer Vision Based Cyclist Detection System

Dec 2025

- Designed and showcased a cyclist detection system in collaboration with Cycle Toronto, leveraging YOLOv8 (transfer learning), Swift-SRGAN, and Arduino to deliver real-time right-hook collision prevention for urban cyclists in Toronto.

SKILLS

- Programming Languages: Proficient in Python, MATLAB, C, and SystemVerilog.
- Design & Modelling Software: Skilled in Fusion, AutoCAD, PSpice, Onshape, SketchUp, and 3D modelling tools.
- Machining: Technical Drawings, Lathe, Mill, Drill Press, Turning, Sawing and Cutting.
- Languages: Fluent in Mandarin, English, and Shanghainese; basic Cantonese.

ACHIEVEMENTS & AWARDS

UTMIST Battle of Schools 2nd Place

University of Toronto - Sept 2026

International Scholar Award & Scholarship

University of Toronto - Sept 2024

UTEK Programming Top 7 Finalist

University of Toronto - Feb 2025